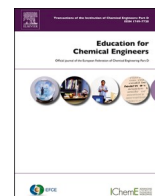




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Editorial: The modern face of chemical engineering in Latin America

In the ever-evolving field of chemical engineering, we find ourselves at a pivotal point. Our profession, once primarily linked to the traditional realms of oil and gas, now spans a vast spectrum of industries, each with its unique challenges and opportunities. From biomedicine to the constantly expanding world of sustainable natural products or customized products by integrated progress in polymer science, colloids science, macroscopic, mesoscopic, and microscopic physics, and advanced instrumental techniques, the chemical engineers are no longer just the behind-the-scenes technicians; they are the vanguard of innovation, the trailblazers charting new territories. Their expanded roles as researchers, innovators, and entrepreneurs are not just a response to the discipline's growth but a reflection of our changing world and the array of opportunities it presents.

With its abundant natural resources and dynamic emerging economies, Latin America stands as a beacon in the global narrative of chemical engineering. The region's unparalleled potential in producing specialty chemicals derived both from traditional and innovative sustainable sources offers a distinct competitive edge in the complex landscape of global manufacturing. This strategic positioning is not merely a geographical or resource-based advantage. It is a confluence of the region's rich history, its vibrant culture of innovation, and a forward-looking vision that aligns with the global push towards greener, more sustainable industries. As the world grapples with environmental challenges and seeks sustainable solutions, Latin America's role becomes even more central. Emphasizing this, there is an immediate call to nurture the upcoming generation of chemical engineers with diverse skills, profound knowledge, and foundational principles that encourage them to drive innovation and sustainability. This holistic development approach will foster a shift from a raw material export-focused region to one that pioneers finished product industries, thereby bridging the socioeconomic disparities prevalent in the area.

In our push towards sustainability, we also need to examine the way we approach chemical engineering education. Modern chemical engineers are not just skilled in formulas; they're comprehensive thinkers, adept at problem-solving, and always at the forefront of innovation. Rather than adding more subjects to an already detailed curriculum, we should blend different areas of knowledge, making learning more connected and relevant. The articles in this issue are proof of these ongoing efforts. They display how educators in Latin America are implementing and living these changes. Our aim is to develop chemical engineers who excel in both specific technicalities and overarching global challenges. As you engage with these pieces, consider the potential of such a comprehensive educational approach and the doors it might open for the future.

However, this leadership in the field of chemical engineering is not

without its challenges. The vast and intricate world of chemical engineering demands not only technical proficiency but also a broader set of skills that are often overlooked in traditional curricula. How do we ensure that our students are not only masters of their craft but also effective communicators, team players, and ethical decision-makers? How can we adapt our educational frameworks to resonate with the unique opportunities and challenges of the Latin American context? These questions are not just theoretical ponderings but pressing concerns that educators grapple with daily.

In response to these challenges, this special issue, titled 'Latest pedagogical developments in chemical engineering education in Latin America: Innovative approaches inside and outside the classroom', highlights the resilience, innovation, and dedication of educators across Latin America. These educators are not merely reacting to change; they are proactively shaping the future of education in our region. Through novel pedagogical interventions, meaningful collaborations with local industries, and a keen focus on addressing societal issues, they are setting new benchmarks and redefining the contours of chemical engineering education for the next generation.

The articles in this collection offer a panoramic view of the educational landscape in chemical engineering across Latin America. Each manuscript, meticulously crafted, delves into specific facets of pedagogical innovation, reflecting the diverse challenges and opportunities unique to the region.

1. Modernizing and refining educational curricula

Rafael Amaya-Gómez and his team explored the intricate relationship between risk perception and decision-making among engineering students in Colombia (Amaya-Gómez, 2023). Their pedagogical approach emphasizes the importance of understanding students' instinctive attitudes toward risk. By employing a survey strategy and a weighted approach following a Factor Analysis, they were able to classify students' tendencies toward risk. This invaluable insight serves as a foundation for refining pedagogical strategies in process safety education, ensuring a more holistic and informed approach to teaching in engineering programs.

At the National University of Colombia, Alvaro Orjuela and colleagues have integrated problem-based methodologies into their Unit Operations Laboratories (Orjuela et al., 2023). This pedagogical approach not only enhances students' technical competencies but also fosters a deeper understanding of core chemical engineering operations. By encouraging students to engage in hands-on projects, they are immersed in an environment that promotes critical thinking, problem-solving, and a genuine passion for research and innovation.

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The University of Campinas in Brazil, as described by Luís Fernando Mercier Franco and his team, has embarked on a transformative journey towards competency-based education (Franco, 2023). Their pedagogical methodology is rooted in active learning, where students are encouraged to engage with real-world challenges, fostering a deeper connection between theoretical knowledge and practical application. This approach ensures that students are not only well-versed in the technical aspects of their discipline but also equipped with the soft skills necessary for effective communication, teamwork, and leadership.

Bruno Ramos and his team highlight the educational modernization process at the Polytechnic School of the University of São Paulo (Ramos, 2023). Their pedagogical initiatives emphasize the importance of content relevance, innovative teaching spaces, and fostering core skills. By integrating these elements, they aim to create a learning environment that resonates with today's students, promoting engagement, critical thinking, and a genuine passion for the discipline.

The approach to teaching safety at Universidad Pontificia Bolivariana, as discussed by another team, is a testament to the importance of instilling a safety culture from the onset of one's educational journey. Their pedagogical strategy, branded as "The moment of safety," integrates safety education throughout the curriculum, ensuring that students are well-equipped to navigate the complexities of the engineering profession with a keen awareness of safety protocols and best practices (Ocampo-López, 2023).

In light of evolving educational paradigms, the Universidad de la República in Uruguay has inaugurated a competency-based Chemical Engineering undergraduate program in 2021, as meticulously articulated by E. Castelló et al (Castelló, 2023). The paper delineates the curriculum restructuring process, juxtaposing the defined competencies against the benchmarks of the Institution of Chemical Engineers. Through three pragmatic cases, it unveils the integration of essential skills, showcasing a strategic pedagogical shift designed to foster real-world competencies, awaiting its first cohort in 2023. This initiative heralds a progressive trajectory in chemical engineering education in Uruguay, aligning with global educational trends.

2. Innovative approaches to teaching and learning

Martín N. Gatti and his team present a novel pedagogical approach to teaching thermodynamics (Gatti and Martín, 2023). By emphasizing the concept of exergy, they aim to foster a deeper understanding of energy forms and their transmission between systems. Their pedagogical proposal integrates various learning axes, ensuring that students are equipped with both the theoretical knowledge and practical skills necessary for meaningful learning in thermodynamics.

The Federal University of São Paulo's Chemical Engineering Undergraduate Program, as described, has integrated a competency-based approach into its curriculum. Their pedagogical strategy emphasizes the importance of transdisciplinary competencies, ensuring that students are well-rounded and equipped to tackle a wide range of challenges in the field of chemical engineering (de Oliveira and Assis, 2023).

Universidad Pontificia Bolivariana in Medellín, Colombia, has employed a project-based learning methodology in its chemical engineering curriculum (Rendón-Castrillón et al., 2023). This pedagogical approach emphasizes the importance of hands-on learning, fostering a deeper connection between theoretical knowledge and practical application. By challenging students to engage in real-world research projects, they are encouraged to think critically, collaborate effectively, and develop a genuine passion for innovation and discovery.

Lastly, Universidad del Valle's chemical engineering program has undergone a significant curriculum reform rooted in a student-centred constructivist approach (Gómez-Ríos et al., 2023). Their pedagogical strategy emphasizes the importance of holistic learning, integrating both

technical knowledge and soft skills into the curriculum. By focusing on the Sensitivities, Capabilities, and Competencies necessary for effective engineering practice, they aim to create a learning environment that resonates with today's students, fostering engagement, critical thinking, and a genuine passion for the discipline.

Together, these contributions not only paint a vivid picture of the current state of chemical engineering education in Latin America but also hint at the exciting directions it could take in the future. They serve as a testament to the region's commitment to excellence, adaptability, and forward-thinking, illuminating the multifaceted endeavours of educators who are tailoring the future of the discipline to meet the specific demands and aspirations of the region.

As we proudly present this curated collection, we extend a heartfelt invitation to our readers. We urge you to immerse yourself in these articles, to critically engage with the diverse perspectives they present, and to reflect deeply on the insights and lessons they offer. This is more than just a collection of writings; it's a conversation, a dialogue about the future of chemical engineering education. As you navigate through these pages, we hope you'll be inspired to join this conversation, contributing your voice and vision to the ongoing discourse about the future of chemical engineering education in Latin America and the broader global context.

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